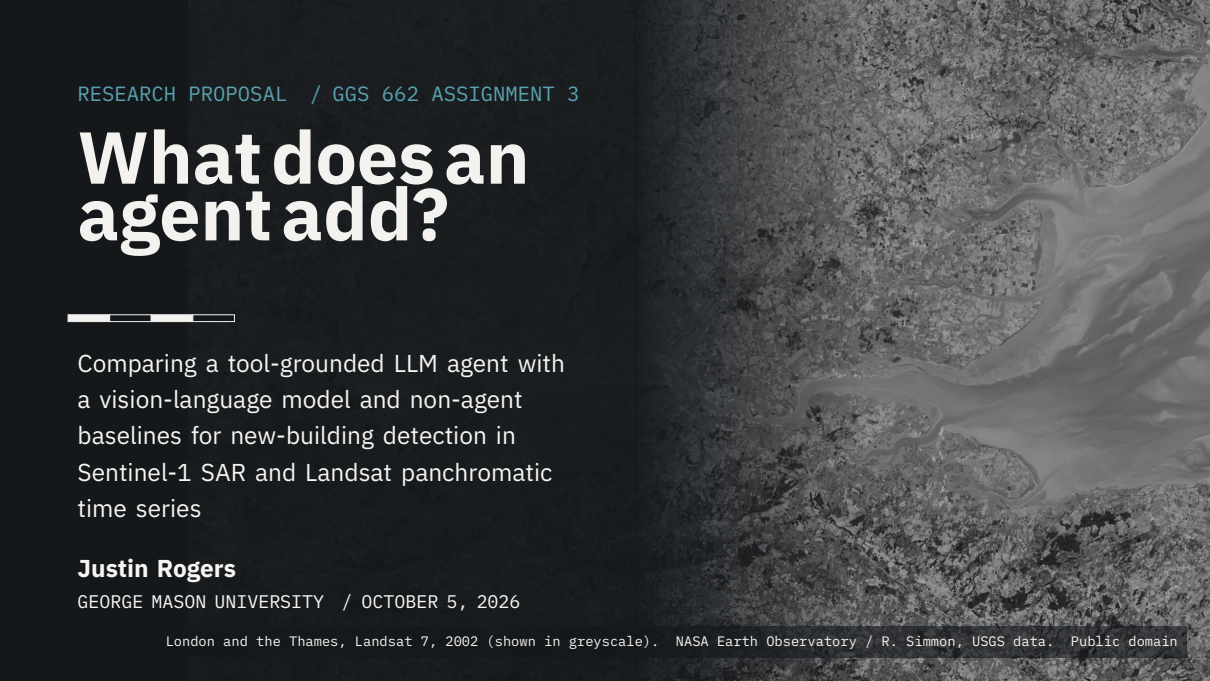
RESEARCH PROPOSAL / GGS 662 ASSIGNMENT 3

What does an agent add?

Comparing a tool-grounded LLM agent with a vision-language model and non-agent baselines for new-building detection in Sentinel-1 SAR and Landsat panchromatic time series

Justin Rogers
GEORGE MASON UNIVERSITY / OCTOBER 5, 2026

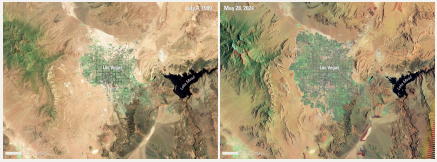


London and the Thames, Landsat 7, 2002 (shown in greyscale). NASA Earth Observatory / R. Simmon, USGS data. Public domain

PROBLEM

The problem: large image archives, unreliable AI interpreters

- Satellite archives now hold years of monthly SAR and optical images per place. Reading change from them by hand does not scale.
- Current multimodal models are weak at this:
  - best of 18 models: 34.1% **accuracy** on a city-change benchmark (Xuan et al., 2025)
  - LLM agents on yes/no Earth-observation questions: 40% **accuracy** zero-shot, about 60% with self-reflection, where guessing scores 50% (Kao et al., 2026)
- Tool-using agents are the promising route (Feng et al., 2026), but SAR and panchromatic data are rarely tested.



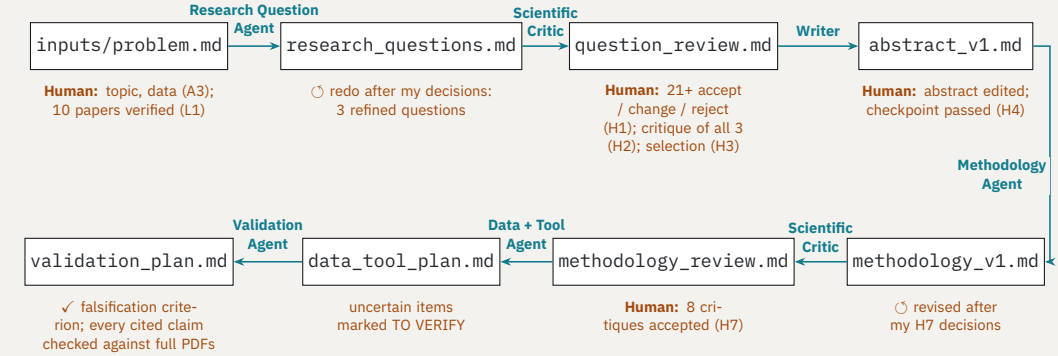
Las Vegas 1999 vs 2024 (Landsat 7, USGS; public domain)

What this study tests

Whether grounding an LLM in sensor tools makes it more accurate and less prone to fabrication than a VLM reading the same imagery, and what its reasoning adds over non-agent rules given the same tool outputs.

WORKFLOW

The course workflow: state → agent → new state



Human = my decision ↻ = revision ✓ = evidence checked Each agent: fresh conversation, packet only.

QUESTIONS

Three refined questions and my critique

Question (origin)	**Scientific value (my critique)**	**Feasibility in 8 weeks (my critique)**	**Outcome**
Q1. Is a tool-using agent *no worse* than a fixed pipeline with the same tools, better where the tools disagree, and does it abstain when data are missing? (AI, edited by me)	Strong comparison. It holds the tools fixed, so it isolates the LLM's reasoning; the tools, not the LLM, do the detection.	Feasible at test level; a production system across many areas would take longer.	**Selected**, then reshaped by me: VLM comparison added, identical inputs
Q2. Does panchromatic imagery (15 m resolution) add anything to Sentinel-1 SAR? (AI, edited by me)	Good idea, but I would flip it: panchromatic supported by SAR, since panchromatic is more common in commercial imagery.	Feasible only as a panchromatic-only vs panchromatic + SAR analysis inside Q1.	**Kept inside Q1** as a secondary analysis, run both ways (panchromatic-led and SAR-led)
Q3. Does neighbouring-cell context help the agent place change in the right cell? (AI)	Reasonable, but too general: it names no specific failure to fix (e.g. radar signal shifted into the wrong cell), so it has no clear hypothesis.	Feasible.	**Deferred** to follow-on work (not rejected)

SELECTION

From three questions to one: Q1, reshaped by my critique

Research question

In held-out SpaceNet 7 areas (2017–2020), does a tool-grounded LLM agent detect new buildings in 100 m cells (presence and onset month) more accurately than a VLM viewing the same imagery and than non-agent baselines given the same tool outputs?

How the three critiques shaped it

- **Q1** isolates what the agent adds (same tools); I added the VLM comparison, my core interest.
- **Q2** lives on inside it: panchromatic-only vs panchromatic + SAR and the reverse, both cheap with the pipeline.
- **Q3** is deferred: reasonable and feasible, but too general for 8 weeks.

Hypothesis

- Tool grounding reduces hallucination: the agent is more accurate than the VLM and fabricates fewer answers.
- Against a tuned rule and a fitted rule on the same tool outputs, the agent is at least as accurate (estimated as a difference with a confidence interval; any margin set from the pilot) and better where the tools disagree (pre-registered definition).

CRITICAL REVIEW

AI suggestions I accepted, changed or rejected

AI suggestion	Decision	Why
Critic: an agent-vs-VLM comparison is confounded; first log dropped the VLM	**Changed**	**Reinstated by me:** it is my core question. Made fair with 1 km chips and the VLM's full frame limit, and paired with same-tool baselines.
Hand-label 300–500 cells; use NAIP for dates	**Rejected**	SpaceNet 7 already has monthly building labels from separate imagery; saves about two weeks.
Critic: add a fitted baseline; replace the margin δ with confidence intervals	**Accepted**	A second baseline on the same tool outputs; intervals instead of an arbitrary margin.
Critic's revised question: agent vs fixed pipeline	**Accepted**	The only contrast that isolates what the agent adds.
Panchromatic as an optional ablation	**Changed**	Required secondary analysis, and panchromatic-led (my Q2 critique).
Single area split into blocks	**Changed**	Whole areas held out (AOI-level split): less leakage, better external validity.
Critic: reject the neighbourhood question (Q3)	**Changed**	Not rejected, just not selected: reasonable and feasible, but too general for this paper; kept as a follow-on.

Full log: 21 decisions (form field H1); earlier outputs and every agent's input packet are archived.

LITERATURE

Literature and contribution

Paper	What it established	What this work adds
Kao et al. (2026), ACL Findings	LLM agents reach about 60% accuracy at best on yes/no EO questions (guessing: 50%); generated code often fails to run	Predefined, deterministic tools instead of free-form code
Feng et al. (2026), ICLR	Tool-using EO agent (Earth-Agent, 104 tools)	A SAR + panchromatic tool set and a same-tools baseline
Xuan et al. (2025), NeurIPS	Best of 18 multimodal models: 34.1% accuracy (questions answered correctly) on a city-change benchmark	Grounds answers in time-series statistics
Lee et al. (2026), IEEE JSTARS	SAR time-series change types; up-steps linked to construction	Uses the SAR change test inside an agent
Qu et al. (2026), Remote Sens. Environ.	Dense time-series change detection; AOI-level split; misalignment creates false change	Same split logic and misregistration checks
Zuo et al. (2026), IEEE JSTARS	Hallucinated change is a known failure; change-vector-analysis calibration plus self-consistency reduces it	Measures abstention and fabricated answers

Proposed contribution: a controlled comparison of a tool-grounded LLM agent with a vision-language model and with same-tool baselines, on a dense SAR + panchromatic archive, scored against independent monthly labels. Proposed, not established.

METHOD

Data and method

Data (all public)

- **Panchromatic:** Landsat-8 band 8 at 15 m resolution (each pixel covers 15 × 15 m of ground), monthly, already co-registered (MuRA-T, 60 SpaceNet 7 areas, 2017–2020)
- **SAR:** Sentinel-1 C-band GRD (10 m pixel spacing, about 20 m resolution), one orbit track per area, co-registered by me
- **Reference:** SpaceNet 7 monthly building footprints, traced on separate PlanetScope imagery
- **Unit:** 100 m cells; positive if $\geq A$ m^2 of new footprint (set from pilot); results by change size; leave-one-area-out over all usable areas

Steps

1. Build tools: SAR change test, panchromatic change magnitude, pretrained CNN change detector, data availability
2. **Tuned baseline:** fixed rule on the same tools, tuned on development areas
3. **Agent:** same tools, answers `presence / onset month / abstain`
4. **VLM:** reads the same 1 km chips on the same 8 dates as the agent's tools
5. **Fitted baseline:** simple classifier on the same tool outputs
6. Controls: same model with tools on vs off
7. Freeze, pre-register, run test areas once

AI-assisted coding: Claude Code writes the data, tool and analysis code; I review every function and test it on synthetic images with known change.

VALIDATION

What we validate, and how

What is validated	How	Fails if
1. The reference labels	Traced on separate PlanetScope imagery; label-lag distribution reported; onset scored within ±1 month	Lag swamps onset error
2. The tools	Tested alone on development areas and on synthetic known change	Tool F1 near chance
3. A fair comparison	Same chips, dates and question logged for agent and VLM	Any input mismatch
4. Evidence, not memory	Opaque cell IDs; same LLM with tools disabled	No-tools accuracy near the agent's
5. The agent uses its tools	Tool evidence shuffled between cells	Accuracy barely drops
6. Less hallucination	Unanswerable queries; same abstain option for VLM; risk–coverage curves; cited values checked	Fabricates as often as the VLM
7. Generalises to new areas	Leave-one-area-out; power sized by area; paired, spatially blocked tests	Agent − VLM interval includes zero (both metrics)
8. Robustness	One-pixel shift, 200 m cells, SAR look direction; 3 repeat runs	Result flips

Metrics: F1 score (balance of precision and recall) and balanced accuracy (co-primary; margin pre-registered after the pilot), precision, recall, onset error in months, fabrication rate (answers given when the correct answer is "abstain"); deterministic scoring, no LLM judge.

Spatial assumptions: sub-pixel co-registration; cells measured in metres; labels mark visible buildings, so SAR change should lead them; SAR layover can shift signal between cells (Hou et al., 2026). **Key risks:** sparse Sentinel-1 scenes in some areas; weak tools at 15 m resolution; VLM setup time; API budget.

PLAN

Eight-week plan, journal and co-authors

Week	Work
1	Download data; choose areas; pick the VLM and document its training data; draft pre-registration
2	Grid, labels, chips; SAR co-registration error; validate tools; **go/no-go gate**
3	Harness for all arms; tune and fit baselines; pilot; set margin, sample size, API cap
4	Pre-register; run all required arms on the test areas
5	Score; few-cluster confidence intervals; robustness
6	Optional extras only if on schedule; draft paper
7	Critic review, claim audit, revise
8	Half-week buffer; package code; submit to JAG

Journal: *International Journal of Applied Earth Observation and Geoinformation*. **2025 JIF 8.2**; source: Clarivate Journal Citation Reports, 2025 JCR year (June 2026 release). Publishes applied EO methods, SAR and building change detection, and GeoAI, which fits an agent-vs-VLM evaluation on SAR + panchromatic time series. **Minimum viable paper:** agent vs fitted baseline vs VLM; robustness checks can be cut.

Possible co-authors (neither has agreed)

- **Dr. Edward Oughton** (course instructor), GMU Geography & Geoinformation Science: agentic GeoAI and spatial computing. Checks: workflow design, evaluation protocol, reproducibility.
- **Dr. Konrad Wessels**, GMU Geography & Geoinformation Science: SAR and time-series remote sensing. Checks: SAR preprocessing, co-registration, physical plausibility of SAR change.

SOURCES

Sources and AI disclaimer

Data: MuRA-T (Purdue RVL; Deshmukh et al., 2023, arXiv:2302.12301, GPL v3); SpaceNet 7 (Van Etten et al., CVPR 2021, CC BY-SA 4.0); Copernicus Sentinel-1 (ESA); Landsat-8 (USGS). **Images:** title, NASA Earth Observatory image by Robert Simmon using USGS Landsat data (public domain, shown in greyscale); slide 2, USGS Landsat Missions, Las Vegas 1999–2024 (public domain). Full references: supporting slides.

AI tools and what they did: Claude Code (Anthropic, Claude Opus 5.5) built the orchestration script and prompt packets; ran all workflow roles as fresh agents (question, critic, writer, methodology, methodology critic, data + tools, validation), with a methodology revision after my H7 decisions, and an independent reviewer agent graded the deck; searched the literature, downloaded the full papers and checked every cited claim against them; found the MuRA-T dataset and openly licensed images; checked the journal metric; drafted the first decision-log rows and data notes at my direction; and generated these slides.

Resources used: Crossref (DOI checks); arXiv, CVF Open Access and publisher PDFs (full papers); Clarivate Journal Citation Reports (journal metric); NASA Earth Observatory and USGS Landsat galleries (images); Purdue RVL and AWS Open Data (datasets).

My own checks: read and approved the 10 study papers (with 2 dataset papers, 12 references); wrote my own critique of the three questions; selected the question and hypothesis; chose the co-authors; added my own decision-log rows after the critics' reviews, including reinstating the VLM comparison, and accepted the methodology critique in H7. Agreement between agents was not treated as validation.

Questions?

WHAT DOES THE AGENT ADD THAT THE SAME TOOLS DO NOT?

SUPPORTING

References (APA 7), 1 of 2

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SUPPORTING

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All 12 full PDFs are saved in the project folder; every claim on these slides was checked against them (`literature/claim_check.md`). Deshmukh et al. (2023) is a preprint; Lee et al. (2026) is an accepted manuscript.

SUPPORTING

Detailed eight-week plan

Week	Tasks	Deliverable	Main risk
1	Download data; area inventory and split; pick the VLM; document VLM training data; draft pre-registration	Inventory; area split	Too few eligible areas
2	100 m grid, reference cells, 1 km chips; Sentinel-1 co-registration error; build and validate core tools; fix minimum footprint area. **Go/no-go:** $\geq$8 test areas and tools pass, else cut scope	Chip set; tool report; gate decision	SAR change test fails at 15 m resolution
3	Harness for agent, VLM, VLM + tool text and controls; tune and fit the two baselines; pilot; area-level power simulation	Pilot results; margin, sample size, API cap	Few conflict cells
4	Pre-register; run all required arms on the test areas	Pre-registration; raw outputs	API limits
5	Score; few-cluster confidence intervals; robustness checks	Results tables and figures	Wide confidence intervals
6	Optional extras (shuffled evidence, CNN, SAR-only) only if on schedule; draft	Full draft	Scope creep
7	Critic review; claim audit; revise	Revised draft	Analysis error found
8	Half-week buffer; package code and data; submit to JAG	Submission	Slippage absorbed

SUPPORTING

An agent output and my revision

Agent (stage 1 redo), Q-B

“Does giving a tool-using LLM agent Landsat-8 panchromatic tools, in addition to Sentinel-1 tools, change how accurately it identifies 100 m cells that gain new buildings?”

Critic (stage 2) and the first decision log

The critic called an agent-vs-VLM comparison confounded; the first decision log (drafted by Claude) dropped the VLM arm.

My revision

Flip the direction: panchromatic is the lead sensor, SAR supports it, because panchromatic is more common in commercial imagery. Kept as a required secondary analysis inside Q1.

My decision

Reinstated the VLM comparison, my core question, and addressed the critique: the VLM gets readable 1 km chips and its full frame limit, and same-tool baselines isolate the agent’s reasoning.

Outputs: `outputs/research_questions.md`, `outputs/question_review.md`; earlier versions and every prompt packet in `outputs/history/` and `outputs/packets_used/`.